

Semester	B.E. Semester VII – Computer Engineering
Subject	Scalable ML & BDA Lab
Subject In-charge	Prof. Shweta Jaiswar

Student Name	Pratik Pujari
Roll Number	23102B0008
Grade and Subject Teacher's Signature	

Experiment Number	02	
Experiment Title	Data Ingestion and ETL Pipeline using PySpark	
Resources / Apparatus Required	Hardware:	Software:
Description	Develop an ETL (Extract, Transform and Load) pipeline using PySpark. Load a real-world dataset, inspect data quality, handle missing values, remove duplicate records, rename and transform columns, create new derived features, and save the cleaned dataset into a new CSV file.	
Observation	- The Netflix dataset contained 8,788 records after data cleaning. - Significant missing values were observed in the director, cast, and country columns, while other columns contained only a few missing entries. - Missing values in non-critical columns were replaced with "Unknown", whereas rows with missing values in essential columns were removed. - No duplicate records were found after performing duplicate analysis. - The release_year column was converted from string to integer to enable numerical operations. - The listed_in column was renamed to genre for improved readability. - Three derived features (content_age, is_movie, and is_tv_show) were created to enrich the dataset and make it more suitable for downstream analysis and machine learning applications.	
Program	<pre>import kagglehub import os from pyspark.sql import SparkSession from pyspark.sql.functions import * spark = SparkSession.builder.appName("Netflix_ETL_Pipeline").getOrCreate() path = kagglehub.dataset_download("sunil123kumar/netflix-movies-and-tv-shows-dataset")</pre>	

	<pre> csv_path = os.path.join(path, "netflix_titles.csv") df = spark.read.csv(csv_path, header=True, inferSchema=True) df = df.fillna({ "director": "Unknown", "cast": "Unknown", "country": "Unknown" }) df = df.dropna(subset=["title", "type", "release_year"]) df = df.dropDuplicates() df = df.withColumn("release_year", col("release_year").cast("integer")) df = df.withColumnRenamed("listed_in", "genre") CURRENT_YEAR = 2026 df = df.withColumn("content_age", lit(CURRENT_YEAR) - col("release_year")) df = df.withColumns({ "is_movie": (col("type") == "Movie").cast("int"), "is_tv_show": (col("type") == "TV Show").cast("int") }) df.coalesce(1).write.mode("overwrite").option("header", True).csv("cleaned_netflix_dataset") </pre>
Output	<pre> root -- show_id: string (nullable = true) -- type: string (nullable = true) -- title: string (nullable = true) -- director: string (nullable = true) -- cast: string (nullable = true) -- country: string (nullable = true) -- date_added: string (nullable = true) -- release_year: string (nullable = true) -- rating: string (nullable = true) -- duration: string (nullable = true) -- listed_in: string (nullable = true) -- description: string (nullable = true) </pre> Schema of the CSV

```

+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|show_id|type|title|director|cast|country|date_added|release_year|rating|duration|listed_in|description|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|      0|  1|   2|    2636| 826|   832|      13|         2|    6|    5|    3|    3|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+

```

Missing values in each colms

```

+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|show_id|type|title|director|cast|country|date_added|release_year|rating|duration|listed_in|description|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|      0|  0|  0|    0|  0|    0|    0|         0|    0|    0|    0|    0|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+

```

After handling null to "Unknown" and dropping some null values if minority

```

root
|-- show_id: string (nullable = true)
|-- type: string (nullable = true)
|-- title: string (nullable = true)
|-- director: string (nullable = false)
|-- cast: string (nullable = false)
|-- country: string (nullable = false)
|-- date_added: string (nullable = true)
|-- release_year: integer (nullable = true)
|-- rating: string (nullable = true)
|-- duration: string (nullable = true)
|-- listed_in: string (nullable = true)
|-- description: string (nullable = true)

```

After handling all Nulls colms were auto inferred to **nullable=false**

```

root
|-- show_id: string (nullable = true)
|-- type: string (nullable = true)
|-- title: string (nullable = true)
|-- director: string (nullable = false)
|-- cast: string (nullable = false)
|-- country: string (nullable = false)
|-- date_added: string (nullable = true)
|-- release_year: integer (nullable = true)
|-- rating: string (nullable = true)
|-- duration: string (nullable = true)
|-- listed_in: string (nullable = true)
|-- description: string (nullable = true)

```

Casted release_yr
to int
from string

title	release_year	content_age	type	Movies	TV Shows
Dick Johnson Is Dead	2020	6	Movie	true	false
Blood & Water	2021	5	TV Show	false	true
Ganglands	2021	5	TV Show	false	true
Jailbirds New Orleans	2021	5	TV Show	false	true
Kota Factory	2021	5	TV Show	false	true
Midnight Mass	2021	5	TV Show	false	true
My Little Pony: A New Generation	2021	5	Movie	true	false
Sankofa	1993	33	Movie	true	false
The Great British Baking Show	2021	5	TV Show	false	true
The Starling	2021	5	Movie	true	false

Created new Features Content age from release_year, and Movies and TV Shows from type

title	Movies	TV Shows
Dick Johnson Is Dead	1	0
Blood & Water	0	1
Ganglands	0	1
Jailbirds New Orleans	0	1
Kota Factory	0	1
Midnight Mass	0	1
My Little Pony: A New Generation	1	0
Sankofa	1	0
The Great British Baking Show	0	1
The Starling	1	0

only showing top 10 rows

Converted the colms from bool to int for OHE

Final OP:

1st Half

show_id	type	title	director	cast	country	date_added
s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	Unknown	United States	September 25, 2021
s2	TV Show	Blood & Water	Unknown	Ama Qamata, Khosi...	South Africa	September 24, 2021
s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tr...	Unknown	September 24, 2021
s4	TV Show	Jailbirds New Orl...	Unknown	Unknown	Unknown	September 24, 2021
s5	TV Show	Kota Factory	Unknown	Mayur More, Jiten...	India	September 24, 2021
s6	TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach...	Unknown	September 24, 2021
s7	Movie	My Little Pony: A...	Robert Cullen, Jo...	Vanessa Hudgens, ...	Unknown	September 24, 2021
s8	Movie	Sankofa	Haile Gerima	Kofi Ghanaba, Oya...	United States, Gh...	September 24, 2021
s9	TV Show	The Great British...	Andy Devonshire	Mel Giedroyc, Sue...	United Kingdom	September 24, 2021
s10	Movie	The Starling	Theodore Melfi	Melissa McCarthy, ...	United States	September 24, 2021
s11	TV Show	Vendetta: Truth, ...	Unknown	Unknown	Unknown	September 24, 2021
s12	TV Show	Bangkok Breaking	Kongkiat Komesiri	Sukollawat Kanaro...	Unknown	September 23, 2021
s13	Movie	Je Suis Karl	Christian Schwochow	Luna Wedler, Jann...	Germany, Czech Re...	September 23, 2021
s14	Movie	Confessions of an...	Bruno Garotti	Klara Castanho, L...	Unknown	September 22, 2021
s15	TV Show	Crime Stories: In...	Unknown	Unknown	Unknown	September 22, 2021
s16	TV Show	Dear White People	Unknown	Logan Browning, B...	United States	September 22, 2021
s17	Movie	Europe's Most Dan...	Pedro de Echave G...	Unknown	Unknown	September 22, 2021
s18	TV Show	Falsa identidad	Unknown	Luis Ernesto Fran...	Mexico	September 22, 2021
s19	Movie	Intrusion	Adam Salky	Freida Pinto, Log...	Unknown	September 22, 2021
s20	TV Show	Jaguar	Unknown	Blanca Suárez, Iv...	Unknown	September 22, 2021

only showing top 20 rows

	2 nd Half																																																																																																																																																																								
	<table><tr><th>release_year</th><th>rating</th><th>duration</th><th>genre</th><th>description</th><th>is_movie</th><th>is_series</th><th>content_age</th></tr><tr><td>2020</td><td>PG-13</td><td>90 min</td><td>Documentaries</td><td>As her father nea...</td><td>1</td><td>0</td><td>6</td></tr><tr><td>2021</td><td>TV-MA</td><td>2 Seasons</td><td>International TV ...</td><td>After crossing pa...</td><td>0</td><td>1</td><td>5</td></tr><tr><td>2021</td><td>TV-MA</td><td>1 Season</td><td>Crime TV Shows, I...</td><td>To protect his fa...</td><td>0</td><td>1</td><td>5</td></tr><tr><td>2021</td><td>TV-MA</td><td>1 Season</td><td>Docuseries, Reali...</td><td>Feuds, flirtation...</td><td>0</td><td>1</td><td>5</td></tr><tr><td>2021</td><td>TV-MA</td><td>2 Seasons</td><td>International TV ...</td><td>In a city of coac...</td><td>0</td><td>1</td><td>5</td></tr><tr><td>2021</td><td>TV-MA</td><td>1 Season</td><td>TV Dramas, TV Hor...</td><td>The arrival of a ...</td><td>0</td><td>1</td><td>5</td></tr><tr><td>2021</td><td>PG</td><td>91 min</td><td>Children & Family...</td><td>Equestria's divid...</td><td>1</td><td>0</td><td>5</td></tr><tr><td>1993</td><td>TV-MA</td><td>125 min</td><td>Dramas, Independe...</td><td>On a photo shoot ...</td><td>1</td><td>0</td><td>33</td></tr><tr><td>2021</td><td>TV-14</td><td>9 Seasons</td><td>British TV Shows,...</td><td>A talented batch ...</td><td>0</td><td>1</td><td>5</td></tr><tr><td>2021</td><td>PG-13</td><td>104 min</td><td>Comedies, Dramas</td><td>A woman adjusting...</td><td>1</td><td>0</td><td>5</td></tr><tr><td>2021</td><td>TV-MA</td><td>1 Season</td><td>Crime TV Shows, D...</td><td>"Sicily boasts a ...</td><td>0</td><td>1</td><td>5</td></tr><tr><td>2021</td><td>TV-MA</td><td>1 Season</td><td>Crime TV Shows, I...</td><td>Struggling to ear...</td><td>0</td><td>1</td><td>5</td></tr><tr><td>2021</td><td>TV-MA</td><td>127 min</td><td>Dramas, Internati...</td><td>After most of her...</td><td>1</td><td>0</td><td>5</td></tr><tr><td>2021</td><td>TV-PG</td><td>91 min</td><td>Children & Family...</td><td>When the clever b...</td><td>1</td><td>0</td><td>5</td></tr><tr><td>2021</td><td>TV-MA</td><td>1 Season</td><td>British TV Shows,...</td><td>Cameras following...</td><td>0</td><td>1</td><td>5</td></tr><tr><td>2021</td><td>TV-MG</td><td>4 Seasons</td><td>TV Comedies, TV D...</td><td>"Students of colo...</td><td>0</td><td>1</td><td>5</td></tr><tr><td>2020</td><td>TV-MA</td><td>67 min</td><td>Documentaries, In...</td><td>Declassified docu...</td><td>1</td><td>0</td><td>6</td></tr><tr><td>2020</td><td>TV-MA</td><td>2 Seasons</td><td>Crime TV Shows, S...</td><td>Strangers Diego a...</td><td>0</td><td>1</td><td>6</td></tr><tr><td>2021</td><td>TV-14</td><td>94 min</td><td>Thrillers</td><td>After a deadly ho...</td><td>1</td><td>0</td><td>5</td></tr><tr><td>2021</td><td>TV-MA</td><td>1 Season</td><td>International TV ...</td><td>In the 1960s, a H...</td><td>0</td><td>1</td><td>5</td></tr></table>	release_year	rating	duration	genre	description	is_movie	is_series	content_age	2020	PG-13	90 min	Documentaries	As her father nea...	1	0	6	2021	TV-MA	2 Seasons	International TV ...	After crossing pa...	0	1	5	2021	TV-MA	1 Season	Crime TV Shows, I...	To protect his fa...	0	1	5	2021	TV-MA	1 Season	Docuseries, Reali...	Feuds, flirtation...	0	1	5	2021	TV-MA	2 Seasons	International TV ...	In a city of coac...	0	1	5	2021	TV-MA	1 Season	TV Dramas, TV Hor...	The arrival of a ...	0	1	5	2021	PG	91 min	Children & Family...	Equestria's divid...	1	0	5	1993	TV-MA	125 min	Dramas, Independe...	On a photo shoot ...	1	0	33	2021	TV-14	9 Seasons	British TV Shows,...	A talented batch ...	0	1	5	2021	PG-13	104 min	Comedies, Dramas	A woman adjusting...	1	0	5	2021	TV-MA	1 Season	Crime TV Shows, D...	"Sicily boasts a ...	0	1	5	2021	TV-MA	1 Season	Crime TV Shows, I...	Struggling to ear...	0	1	5	2021	TV-MA	127 min	Dramas, Internati...	After most of her...	1	0	5	2021	TV-PG	91 min	Children & Family...	When the clever b...	1	0	5	2021	TV-MA	1 Season	British TV Shows,...	Cameras following...	0	1	5	2021	TV-MG	4 Seasons	TV Comedies, TV D...	"Students of colo...	0	1	5	2020	TV-MA	67 min	Documentaries, In...	Declassified docu...	1	0	6	2020	TV-MA	2 Seasons	Crime TV Shows, S...	Strangers Diego a...	0	1	6	2021	TV-14	94 min	Thrillers	After a deadly ho...	1	0	5	2021	TV-MA	1 Season	International TV ...	In the 1960s, a H...	0	1	5
release_year	rating	duration	genre	description	is_movie	is_series	content_age																																																																																																																																																																		
2020	PG-13	90 min	Documentaries	As her father nea...	1	0	6																																																																																																																																																																		
2021	TV-MA	2 Seasons	International TV ...	After crossing pa...	0	1	5																																																																																																																																																																		
2021	TV-MA	1 Season	Crime TV Shows, I...	To protect his fa...	0	1	5																																																																																																																																																																		
2021	TV-MA	1 Season	Docuseries, Reali...	Feuds, flirtation...	0	1	5																																																																																																																																																																		
2021	TV-MA	2 Seasons	International TV ...	In a city of coac...	0	1	5																																																																																																																																																																		
2021	TV-MA	1 Season	TV Dramas, TV Hor...	The arrival of a ...	0	1	5																																																																																																																																																																		
2021	PG	91 min	Children & Family...	Equestria's divid...	1	0	5																																																																																																																																																																		
1993	TV-MA	125 min	Dramas, Independe...	On a photo shoot ...	1	0	33																																																																																																																																																																		
2021	TV-14	9 Seasons	British TV Shows,...	A talented batch ...	0	1	5																																																																																																																																																																		
2021	PG-13	104 min	Comedies, Dramas	A woman adjusting...	1	0	5																																																																																																																																																																		
2021	TV-MA	1 Season	Crime TV Shows, D...	"Sicily boasts a ...	0	1	5																																																																																																																																																																		
2021	TV-MA	1 Season	Crime TV Shows, I...	Struggling to ear...	0	1	5																																																																																																																																																																		
2021	TV-MA	127 min	Dramas, Internati...	After most of her...	1	0	5																																																																																																																																																																		
2021	TV-PG	91 min	Children & Family...	When the clever b...	1	0	5																																																																																																																																																																		
2021	TV-MA	1 Season	British TV Shows,...	Cameras following...	0	1	5																																																																																																																																																																		
2021	TV-MG	4 Seasons	TV Comedies, TV D...	"Students of colo...	0	1	5																																																																																																																																																																		
2020	TV-MA	67 min	Documentaries, In...	Declassified docu...	1	0	6																																																																																																																																																																		
2020	TV-MA	2 Seasons	Crime TV Shows, S...	Strangers Diego a...	0	1	6																																																																																																																																																																		
2021	TV-14	94 min	Thrillers	After a deadly ho...	1	0	5																																																																																																																																																																		
2021	TV-MA	1 Season	International TV ...	In the 1960s, a H...	0	1	5																																																																																																																																																																		
Conclusion:	- In this experiment, an ETL (Extract, Transform, Load) pipeline was successfully implemented using PySpark on the Netflix Movies and TV Shows dataset. - The dataset was extracted from Kaggle, loaded into a Spark DataFrame, and explored to understand its structure and quality. Data preprocessing techniques such as handling missing values, checking for duplicate records, renaming columns, and converting data types were performed to improve data quality. - Additionally, new derived features including <i>content_age</i>, <i>is_movie</i>, and <i>is_tv_show</i> were created to enhance the dataset for future analytical and machine learning tasks. - Finally, the cleaned and transformed dataset was prepared for further analysis, demonstrating how PySpark can efficiently perform scalable data ingestion and ETL operations on real-world datasets.																																																																																																																																																																								